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Uncovering Understanding–Generation Synergy in Unified Multimodal Models

Joint Training Helps Both Tasks — But Only When Visual Computation Is Architecturally Decoupled by Objective

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Unified multimodal models (UMMs) jointly perform visual understanding and image generation within a single model. Do the two objectives help each other, compete, or merely coexist? This paper answers through a three-level investigation — representation, task, and system — in a controlled setting without pretrained vision priors. The verdict: synergy is real at the representation level, but forces both objectives through the same computation path and one will dominate. A task-decoupled architecture that specialises conflicting visual computation while preserving shared semantic interaction avoids this failure mode and yields consistent gains on both objectives simultaneously.

AT A GLANCE

Problem: Does joint understanding/generation training help, hurt, or have no effect when both objectives share the same model?

Why it matters: Unified models promise efficiency, but training instability and objective conflict limit their practical quality.

Method: Three-level analysis (representation, task, system) with task-decoupled architectural fix.

Result: Decoupled architecture improves understanding by 3.2% and generation FID by 21% vs. naive joint baseline.

Evidence: Controlled experiments at 300M–3B parameters; VQA, image captioning, and text-to-image generation benchmarks.

The Big Picture

Large vision-language models have converged on two paradigms: dedicated understanding models (e.g. LLaVA, InternVL) and dedicated generation models (e.g. Stable Diffusion, DALL-E). Unifying both in a single model is appealing: shared representations could reinforce each other, reduce model count in deployment, and enable joint reasoning over perception and synthesis.

Yet practitioners have observed that naive joint training frequently degrades one or both tasks compared to separate specialist models. The reasons are poorly understood. This paper systematically characterises the conflict and proposes a structural solution.

Why Existing Approaches Fall Short

Existing unified models (e.g. Emu, Show-o, Janus) attempt joint training but either (a) use a pretrained vision encoder that introduces its own inductive bias and masks the training interaction, or (b) report mixed results with no theoretical explanation. Three failure modes emerge:

Objective dominance. When both tasks share the same visual encoding layers, the task with stronger or more numerous gradients suppresses the other — whichever has lower loss tends to pull parameters away from the other objective's optimal region.

Representation conflict. Optimal representations for understanding encode semantic categories and attributes; optimal representations for generation encode low-level textures and pixel statistics. Forcing both through the same encoder causes a compromise that is suboptimal for both.

Confounded benchmarks. Papers using pretrained encoders cannot isolate the effect of joint training from the encoder's pretrained features.

Core Mechanism

The paper identifies that conflict arises in *early visual encoding components* (tokeniser, first few attention layers) but not in *semantic interaction layers* (cross-attention, late transformer blocks).

This motivates the task-decoupled architecture:

- **Specialised early encoders:** separate but lightweight visual tokeniser branches for understanding (Enc_u) and generation (Enc_g).
- **Shared semantic backbone:** a single transformer stack processes fused semantic representations from both branches.

Decoupling allows each encoder to optimise its own visual features while the shared backbone captures cross-modal semantics that benefit both tasks.

Technical Formulation

4.1 Joint Training Objective

The standard (conflict-prone) joint objective is:

$$\mathcal{L}_{\text{joint}} = \lambda_u \mathcal{L}_{\text{understanding}} + \lambda_g \mathcal{L}_{\text{generation}}$$

where $\mathcal{L}_u$ is cross-entropy over VQA/captioning tokens and $\mathcal{L}_g$ is a denoising diffusion loss or autoregressive token loss.

4.2 Dominance Score

To quantify asymmetric degradation, the dominance score D is:

$$D = \frac{\text{Acc}_u^{\text{joint}}/\text{Acc}_u^{\text{separate}}}{\text{Acc}_g^{\text{joint}}/\text{Acc}_g^{\text{separate}}}$$

$D \gg 1$: understanding dominates. $D \ll 1$: generation dominates. Naive joint training produces $D \neq 1$, confirming dominance. The task-decoupled model produces $D \approx 1$.

4.3 Decoupled Forward Pass

Given image x and text tokens t:

$$z_u = \text{Enc}_u(x) \quad (1)$$
$$z_g = \text{Enc}_g(x) \quad (2)$$
$$z_{\text{shared}} = \text{TransformerShared}(z_u, z_g, t) \quad (3)$$
$$\mathcal{L}_{\text{decoupled}} = \mathcal{L}_u(z_{\text{shared}}) + \mathcal{L}_g(z_{\text{shared}}) \quad (4)$$

Both encoders see the shared backbone's gradients, enabling beneficial cross-task signal, while their independent parameters specialise without conflict.

4.4 Representation-Level Synergy (CKA)

Synergy is measured via Centred Kernel Alignment (CKA) between understanding-task-optimal and generation-task-optimal representations. After decoupled training, CKA of the encoder outputs is lower (more specialised), but CKA of the semantic backbone outputs is higher (more shared) — confirming specialisation at the encoder and consolidation at the semantic level.

Learning or Inference Procedure

Training follows standard autoregressive pretraining on interleaved image-text data, with the following modification:

1. Separate learnable visual tokenisers are initialised from a shared initialisation but trained independently.
2. Gradients from $\mathcal{L}_u$ update Enc_u and the shared backbone; gradients from $\mathcal{L}_g$ update Enc_g and the shared backbone.
3. The shared backbone receives signal from both tasks at every step.

At inference, only the relevant encoder branch is activated: understanding queries use Enc_u; generation queries use Enc_g.

What the Guarantee Says

The paper proves that under mild smoothness conditions on the loss landscape, the decoupled architecture has a strictly smaller gradient conflict norm than the fully shared architecture:

$$\|\nabla_\theta \mathcal{L}_u + \nabla_\theta \mathcal{L}_g\|^2_{\text{decoupled}} < \|\nabla_\theta \mathcal{L}_u + \nabla_\theta \mathcal{L}_g\|^2_{\text{shared}}$$

whenever the encoder Jacobians for understanding and generation are non-parallel — which holds whenever the two tasks require different visual features, a natural assumption.

Experimental Findings

Understanding benchmarks (VQA, image classification): The task-decoupled model achieves 3.2% average relative improvement over the best single-task baseline and 5.8% over the naive joint model.

Generation benchmarks (FID on COCO, 256×256):

Model	FID↓	Params
Generation-only specialist	7.4	1B
Naive joint UMM	10.3	1B
Task-decoupled (ours)	**8.1**	1.1B

Parameter overhead: The two specialised encoders add 12% extra parameters vs. the fully shared model, but outperform it on both tasks simultaneously.

Ablations and Interpretation

Which layers to specialise? Specialising only the visual tokeniser (earliest layers) provides the largest gain. Specialising all layers including late transformer blocks hurts by breaking shared semantic grounding.

Loss weight sensitivity (λ_u, λ_g): The naive joint model is highly sensitive to loss weighting: accuracy fluctuates by up to 8% as λ_u/λ_g varies from 0.5 to 2.0. The decoupled model is robust across this range.

Scaling (300M, 1B, 3B): The performance gains persist at all three scales, suggesting the conflict is a structural, not a capacity, issue.

CKA trajectories: During training, understanding and generation encoder representations diverge (lower cross-task CKA), while semantic backbone representations converge (higher cross-task CKA) — matching the specialisation-then-fusion theory.

Reference

Penghao Wu, Haiwen Diao, Weichen Fan, Lewei Lu, Dahua Lin, and Ziwei Liu. **Uncovering Understanding-

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